



Ian Goodfellow @goodfellow_ian · Jan 15, 2019

4.5 years of GAN progress on face generation. arxiv.org/abs/1406.2661
arxiv.org/abs/1511.06434 arxiv.org/abs/1606.07536
arxiv.org/abs/1710.10196 arxiv.org/abs/1812.04948



2014



2015



2016



2017



2018

2014 Black and white



2018

Colored
photo-realistic

GANs Over Time

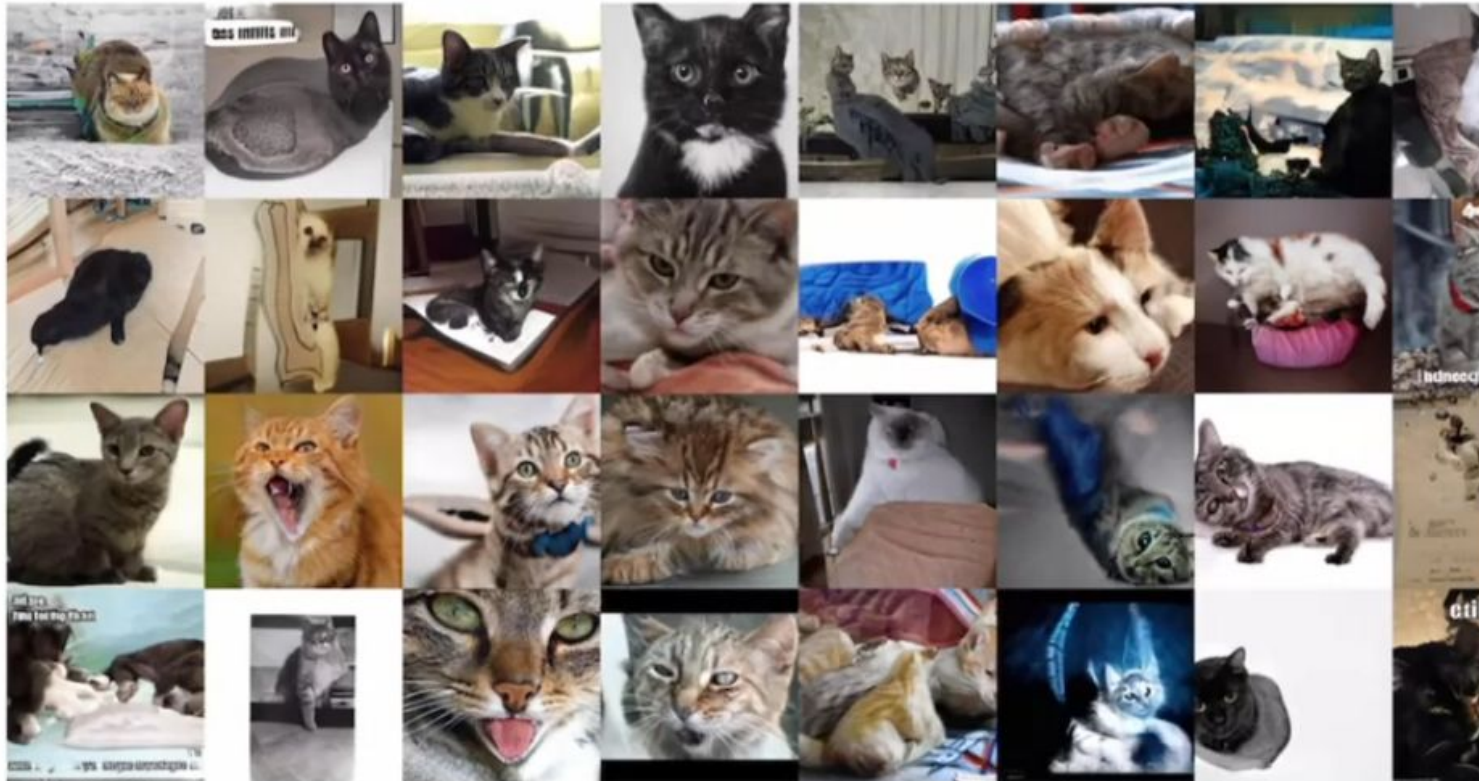


Face Generation
StyleGAN2

These people do
not exist!

Karras, Tero, et al. "Analyzing and improving the image quality of stylegan." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2020.

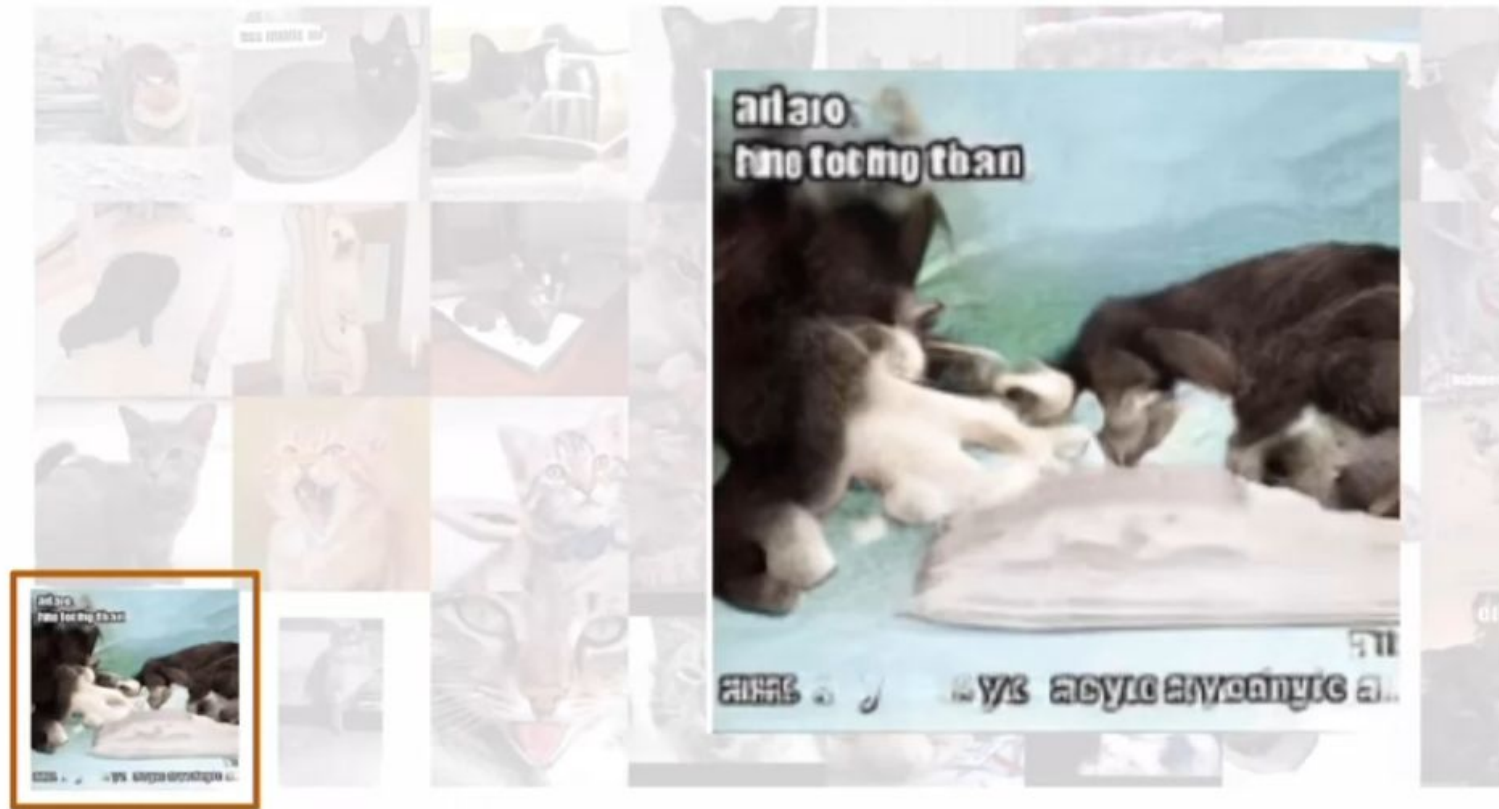
GANs Over Time



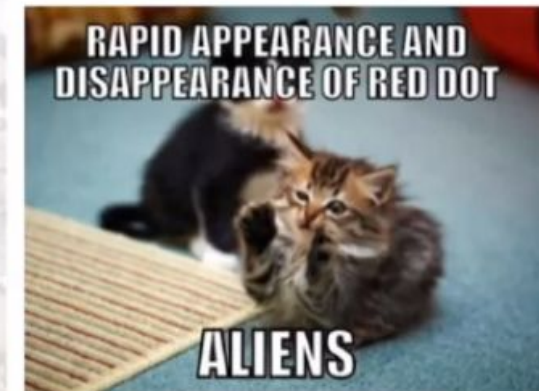
StyleGAN2

Karras, Tero, et al. "Analyzing and improving the image quality of stylegan." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2020.

GANs Over Time



StyleGAN2



Mimics the
distribution of
the training data

Karras, Tero, et al. "Analyzing and improving the image quality of stylegan." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2020.

<https://9gag.com/gag/aWYZKWx>

GANs for Image Translation

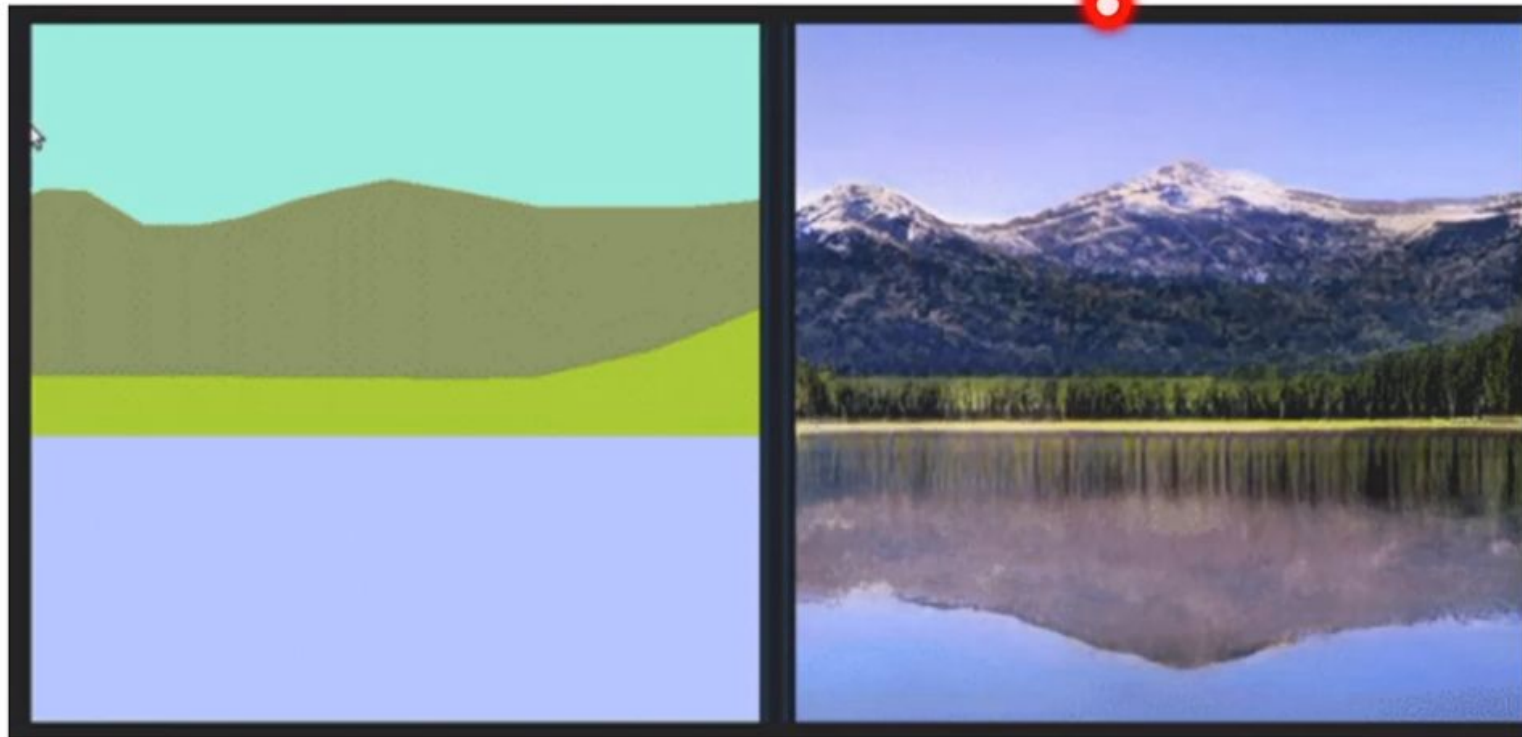
From one domain to
another

CycleGAN



Park, Taesung, et al. "Semantic image synthesis with spatially-adaptive normalization." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2019.

GANs for Image Translation



GauGAN

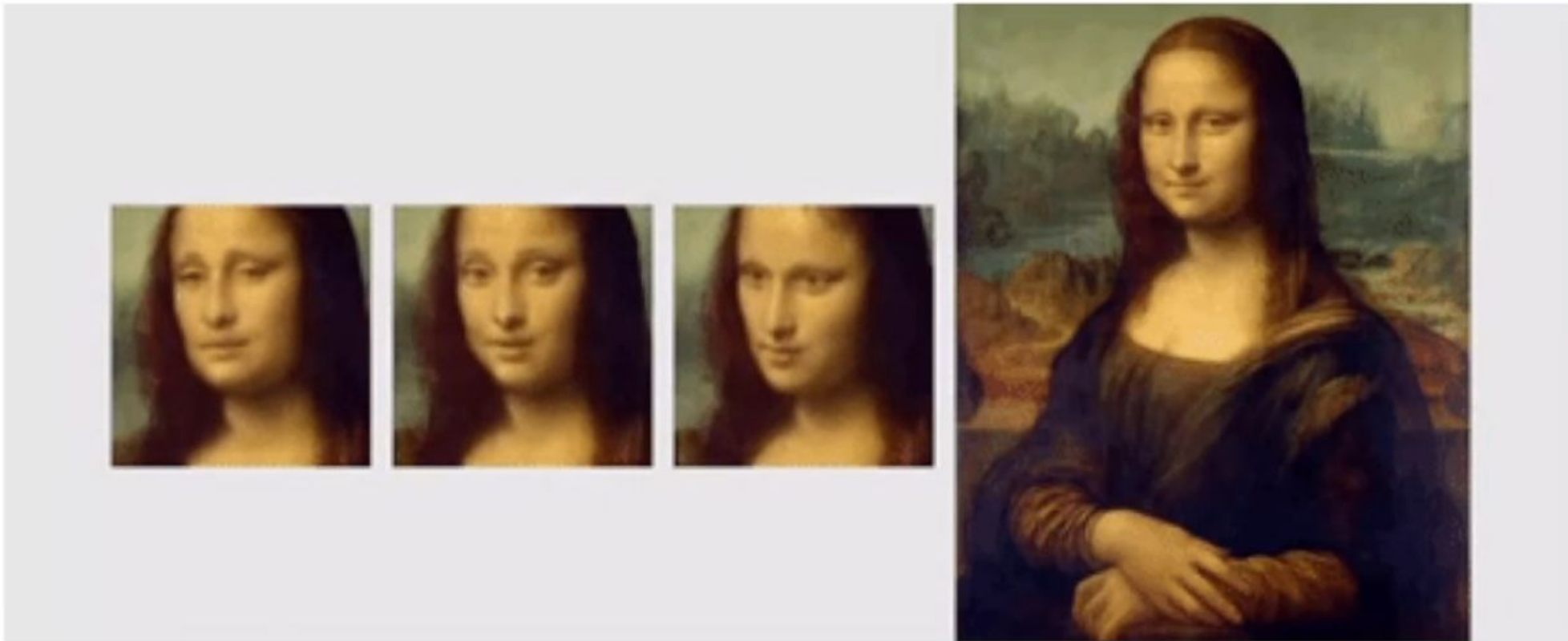
Doodles



Pictures

Park, Taesung, et al. "Semantic image synthesis with spatially-adaptive normalization." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2019.

GANs are Magic!



Zakharov, Egor, et al. "Few-shot adversarial learning of realistic neural talking head models." *Proceedings of the IEEE International Conference on Computer Vision*. 2019.

GANs for 3D Objects



Wu, Jiajun, et al. "Learning a probabilistic latent space of object shapes via 3d generative-adversarial modeling." *Advances in neural information processing systems*. 2016.

Companies Using GANs



Next-gen
Photoshop



Text
Generation



Data
Augmentation

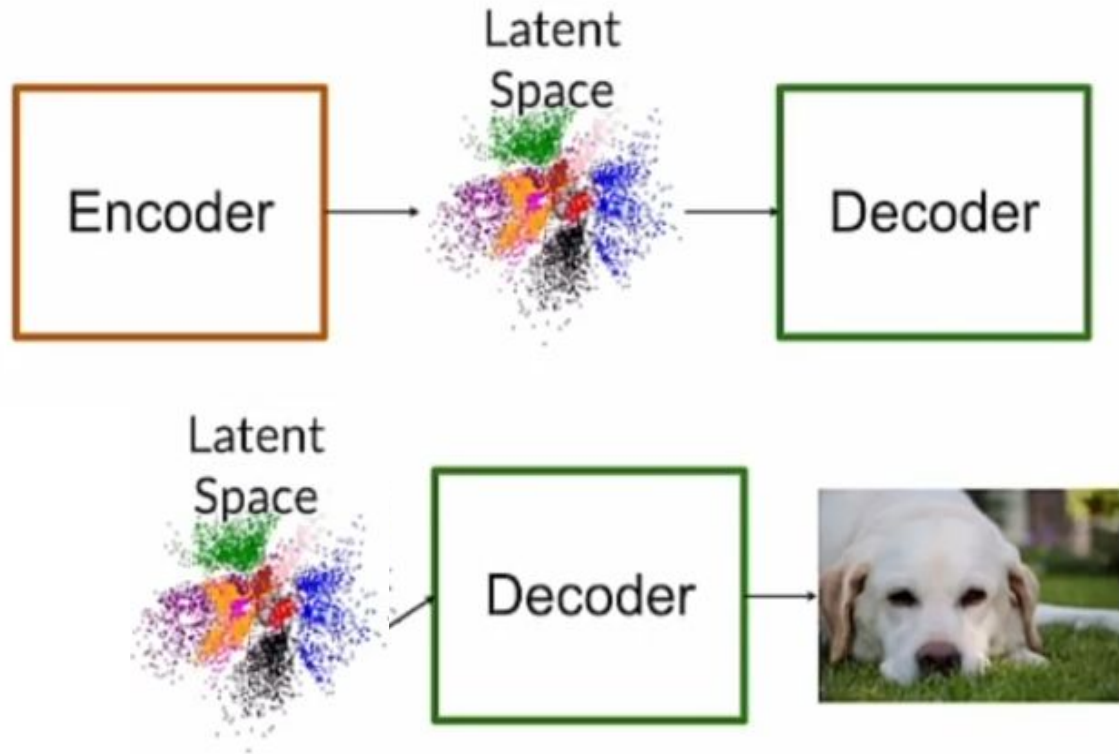


Image Filters

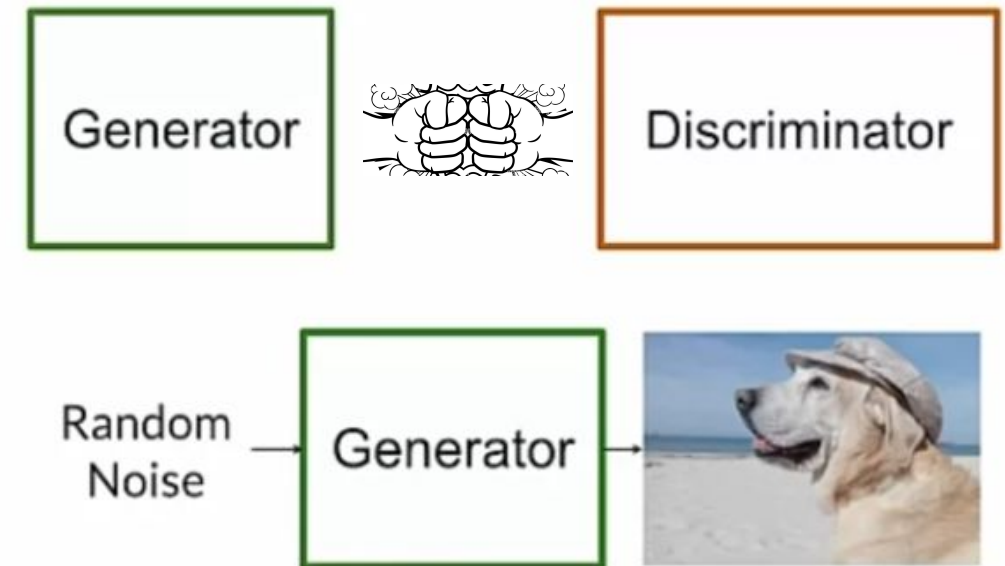


Super-resolution

Variational Autoencoders

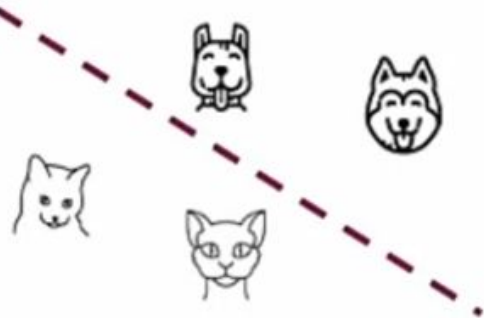


Generative Adversarial Network



Generative Models vs. Discriminative Models

Discriminative models



Features Class

$$X \rightarrow Y$$

$$P(Y|X)$$

Generative models



Noise Class Features

$$\xi, Y \rightarrow X$$

$$P(X|Y)$$

Generative Models vs. Discriminative Models



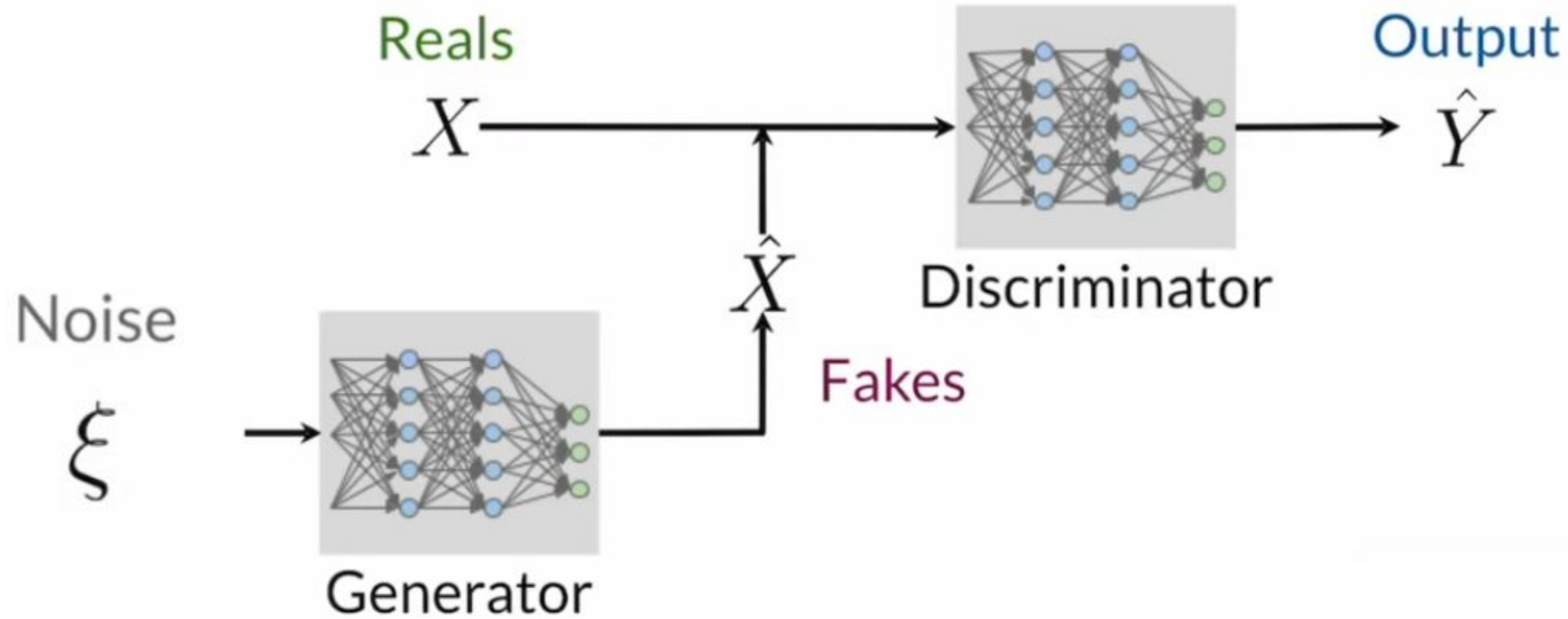
Generative models

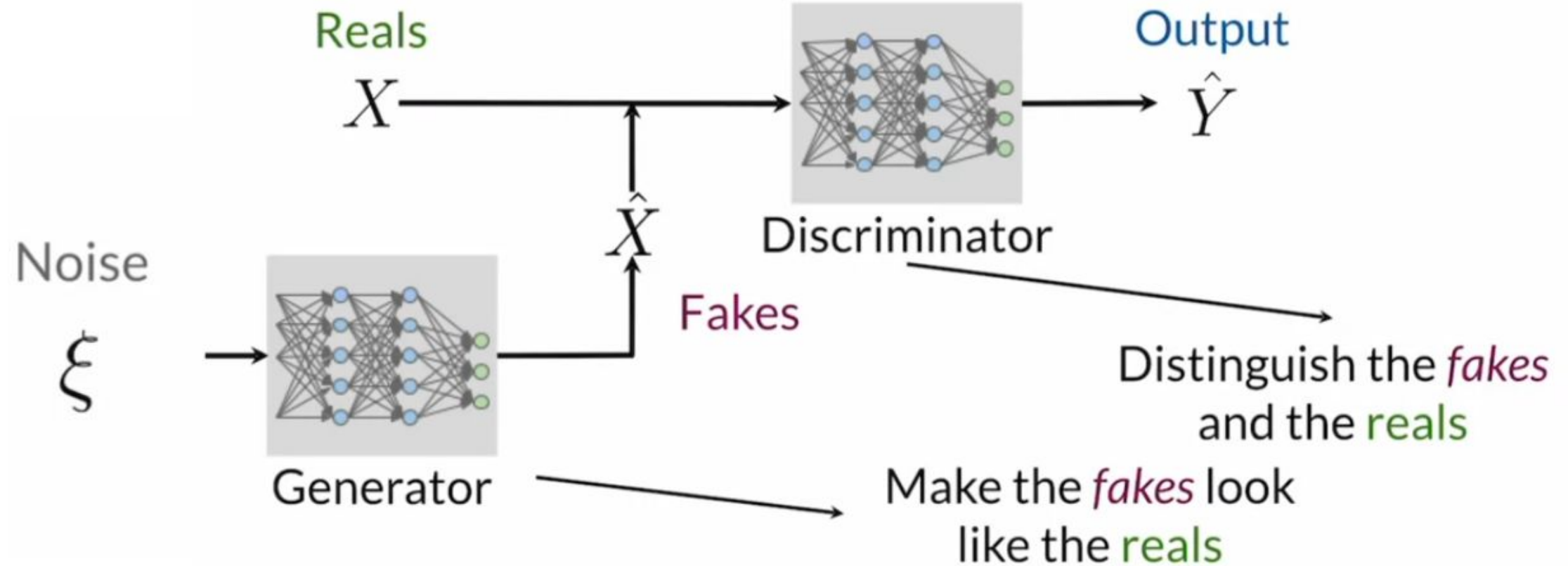


Noise Class Features

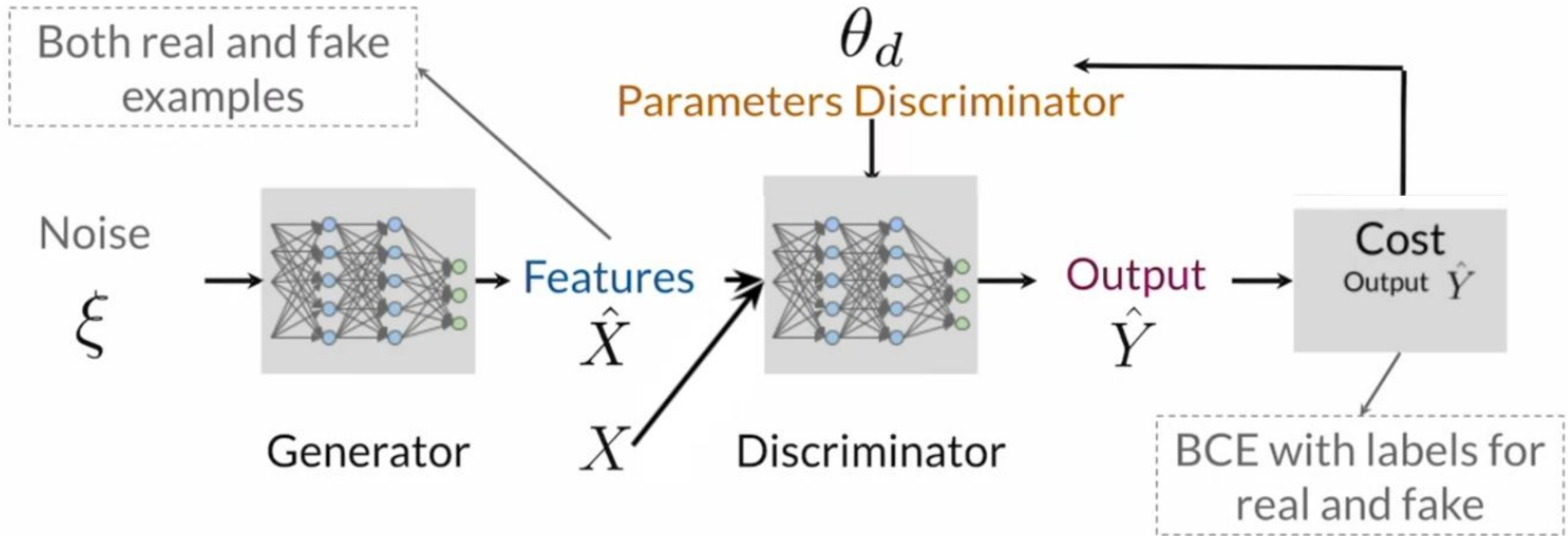
$$\xi, Y \rightarrow X$$
$$P(X|Y)$$

GAN Architecture

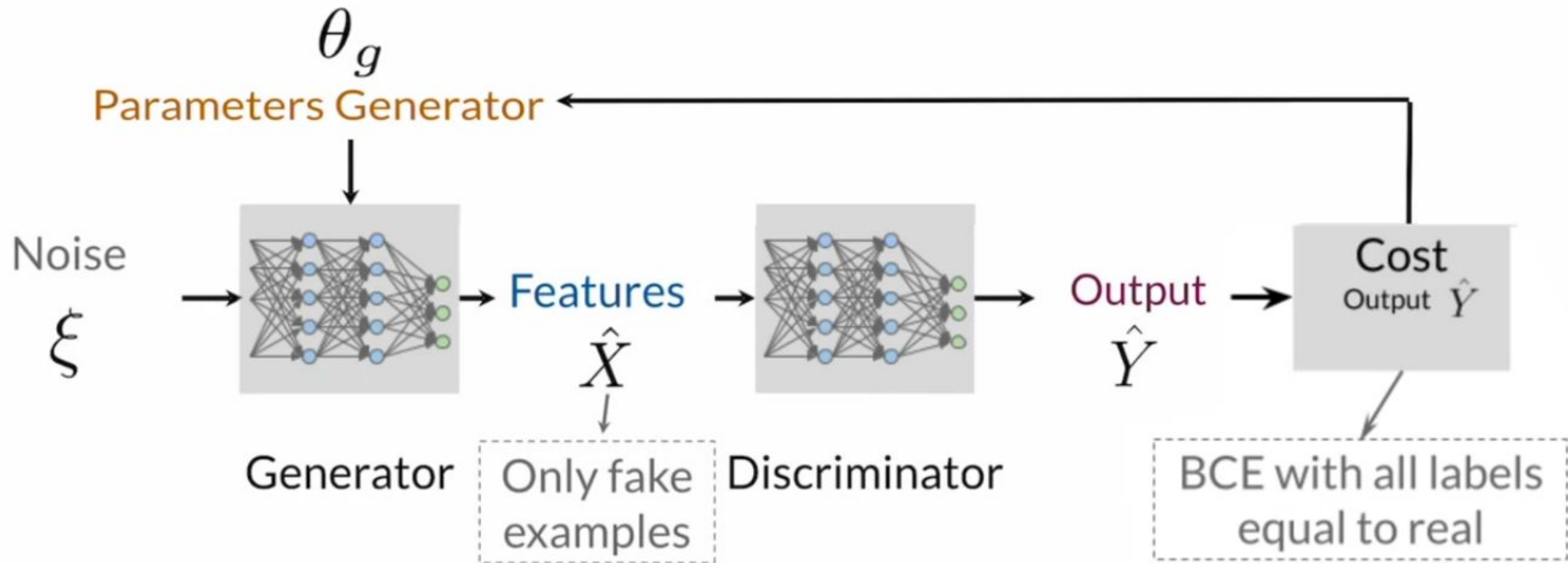




Training GANs: Discriminator



Training GANs: Generator



Superior Discriminator Problem!



- Both model need to improve together.
- And should be kept at similar skill level from the begging of the training.

Deconvolution and Checkerboard Artifacts

AUGUSTUS ODENA Google Brain	VINCENT DUMOULIN Université de Montréal	CHRIS OLAH Google Brain	Oct. 17 2016	Citation: Odena, et al., 2016
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When we look very closely at images generated by neural networks, we often see a strange checkerboard pattern of artifacts. It's more obvious in some cases than others, but a large fraction of recent models exhibit this behavior.

